
DataScience Lab Project

ADVERSARIAL TRAINING ON CIFAR10

ABEL DOUZAL
École normale supérieure PSL
abel.douzal@ens.fr

MIKA LIAO
École normale supérieure PSL
mika.liao@ens.fr

RAYAN DAKHLAOUI
Dauphine PSL
rayane.dakhlaoui@dauphine.eu

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1 Introduction

Deep neural networks achieve impressive performance on image classification tasks, yet remain highly vulnerable to adversarial perturbations. In this project, we study adversarial robustness on CIFAR-10 by reproducing and analyzing established methods rather than proposing new ones. Our work builds directly on the techniques presented in the course, with a particular emphasis on two practical components: the impact of *strong data augmentation* on clean performance, and the effectiveness of the *TRADES* objective Zhang et al. [2019] in improving robustness.

We begin by training a lightweight convolutional network designed for fast inference, which is essential for iterative attacks such as PGD. Extensive data augmentation is employed to improve generalization before introducing adversarial training. We then implement several standard gradient-based attacks to evaluate robustness and to generate adversarial examples during training. Finally, we compare two adversarial training objectives: classical PGD-based training and the TRADES formulation, which explicitly balances natural accuracy and robustness through a KL-regularized loss.

Overall, this report provides an implementation-driven study of how data augmentation and TRADES influence both clean and adversarial performance, offering practical insight into the trade-offs involved in training robust models.

2 Models and Training

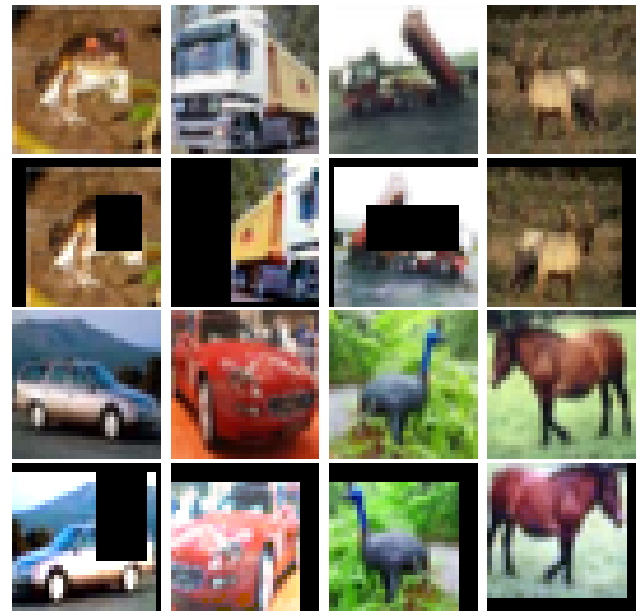
2.1 Deep CNN with Data Augmentation

Model

Our primary objective is to identify a classifier with strong clean accuracy. Since each PGD iteration requires a full forward pass, inference must be as fast as possible. We also cannot rely on pre-computation or transfer-learning shortcuts, because adversarial examples must be generated dynamically during training. The final model we choose is a standard 3-block deep CNN with batch norm and dropout after each layer followed by a fully connected classifier block with dropout.

Data Augmentation

Given the limited size of CIFAR-10, strong generalization further requires extensive data augmentation. This aggressive augmentation may also provide incidental robustness by injecting randomness into the training inputs. For the training, transformations consist of a random crop, a random horizontal flip, slight color jitter, a data augmentation policy from Cubuk et al. [2019] and a final random erasing.



Examples of CIFAR-10 inputs before (top) and after (bottom) data augmentation.

2.2 PGD and FGSM Attacks

Choice of Hyperparameters

To evaluate the robustness of our models and generate adversarial inputs during training, we first consider the simplest gradient-based attacks and follow the hyperparameter recommendations of Madry et al. [2018]. While we directly reuse their optimal step sizes α and perturbation radii ϵ for

both ℓ_2 and ℓ_∞ attacks, we observe markedly different behavior for ℓ_2 attacks on our model. In particular, to make the ℓ_2 attack even moderately effective, we must use a step size of $\alpha = \frac{\epsilon}{4}$ instead of the usual $\alpha = \frac{\epsilon}{\text{nb_steps}}$. Moreover, matching the effectiveness of the ℓ_∞ attack requires scaling the ℓ_2 perturbation budget by almost an order of magnitude, namely $\epsilon = \frac{80}{255}$.

This phenomenon is expected: the ratio between the volumes of the ℓ_∞ ball and the ℓ_2 ball in dimension d grows exponentially with d , since

$$\frac{\text{Vol}(B_\infty^d(1))}{\text{Vol}(B_2^d(1))} = \frac{2^d}{\frac{\pi^{d/2}}{\Gamma(\frac{d}{2}+1)}} \sim \left(\frac{2\sqrt{d}}{\sqrt{2\pi e}} \right)^d \text{ as } d \rightarrow \infty.$$

Thus, the ℓ_∞ ball covers a dramatically larger region of the input space as the dimension increases, making equal values of ϵ incomparable across norms. Scaling ϵ by a factor of 10 for ℓ_2 therefore only partially compensates for this geometric discrepancy.

Example of an Attack on a Non-Robust Model

We illustrate in 1 how our non-robust model reacts to several common adversarial attacks. The clean image is classified correctly with high confidence, but even relatively small perturbations—bounded by $\epsilon = 8/255$ for ℓ_∞ attacks and comparable ℓ_2 budgets—are sufficient to change the prediction entirely. In all cases except FGSM- ℓ_2 , the adversarial perturbation not only causes a misclassification but also increases the model’s confidence in the wrong label.

Although the perturbation is clearly visible for this particular example, it should not have this big of an effect on the model’s prediction. Despite its visibility, this attack remains a strong adversary and therefore a valuable component for adversarial training.

2.3 Adversarial Training

Adversarial training was first popularized by Goodfellow et al. [2015], who proposed generating adversarial examples using the Fast Gradient Sign Method (FGSM) and training the model jointly on clean and perturbed inputs. Formally, the goal is to solve a robust optimization problem of the form

$$\min_{\theta} \mathbb{E}_{(x,y)} \left[\max_{\|\delta\|_p \leq \epsilon} \mathcal{L}(f_{\theta}(x + \delta), y) \right],$$

where the inner maximization constructs an adversarial example within an ℓ_p -ball around x . This formulation motivates replacing FGSM with stronger attacks such as Projected Gradient Descent (PGD), which provides a closer approximation of the worst-case perturbation.

Standard Adversarial Loss. In our first approach, each clean sample x is augmented with a PGD adversarial counterpart x^{adv} , and training minimizes the standard cross-entropy

on the adversarial inputs:

$$\mathcal{L}_{\text{adv}}(\theta) = \mathbb{E}_{(x,y)} [\mathcal{L}(f_{\theta}(x^{\text{adv}}), y)].$$

This corresponds to directly optimizing the robust objective using PGD to approximate the inner maximization.

TRADES Loss. We also evaluate the TRADES loss Zhang et al. [2019], which decomposes robustness into a natural accuracy term and a robustness regularization term using KL divergence. The objective is

$$\mathcal{L}_{\text{TRADES}}(\theta) = \mathbb{E}_{(x,y)} \left[\underbrace{\mathcal{L}(f_{\theta}(x), y)}_{\text{natural accuracy}} + \beta \cdot \underbrace{\text{KL}(f_{\theta}(x) \parallel f_{\theta}(x^{\text{adv}}))}_{\text{robustness penalty}} \right]$$

where β controls the trade-off between standard accuracy and adversarial robustness. TRADES is known to yield smoother and more stable training compared to classical adversarial training.

2.4 Wavelets for feature extraction

We focused on wavelet transform as a feature extractor, with a view to achieving very high robustness to gradient-based attacks and even more general attacks. The main reference for wavelets is Mallat [1999].

Wavelet Definition A wavelet is a localized oscillatory function that acts as a band-pass filter and can be dilated or rotated to probe a signal at multiple scales and orientations. Wavelet analysis relies on a scaling function $\phi \in L^2(\mathbb{R})$ (low-pass filter) and a mother wavelet $\psi \in L^2(\mathbb{R})$ (band-pass filter).

- ψ is localized and oscillatory, satisfying

$$\int_{\mathbb{R}} \psi(x) dx = 0$$

- The family of wavelet atoms $\psi_{j,k}$ is generated by dilation (j : scale) and translation (k : position):

$$\psi_{j,k}(x) = 2^{-j/2} \psi(2^{-j}x - k), \quad (j, k) \in \mathbb{Z}^2.$$

Wavelets such as Daubechies 4 (db4) or Haar are not differentiable. But Meyer’s wavelet is $\mathcal{C}^\infty$.

For 2D signals like images, the basis is obtained via the tensor product of 1D functions:

$$\Phi(\mathbf{x}) = \phi(x_1)\phi(x_2)$$

$$\Psi^H(\mathbf{x}) = \phi(x_1)\psi(x_2)$$

$$\Psi^V(\mathbf{x}) = \psi(x_1)\phi(x_2)$$

$$\Psi^D(\mathbf{x}) = \psi(x_1)\psi(x_2)$$

where Φ is the 2D scaling function and $\Psi^{H,V,D}$ are 2D wavelets capturing horizontal, vertical, and diagonal details. Wavelet atoms are generated by dilation and translation $\mathbf{k} = (k_1, k_2)$ with normalization 2^{-j} :

$$\Psi_{j,\mathbf{k}}^*(\mathbf{x}) = 2^{-j} \Psi^*(2^{-j}\mathbf{x} - \mathbf{k}), \quad \mathbf{x} \in \mathbb{R}^2.$$

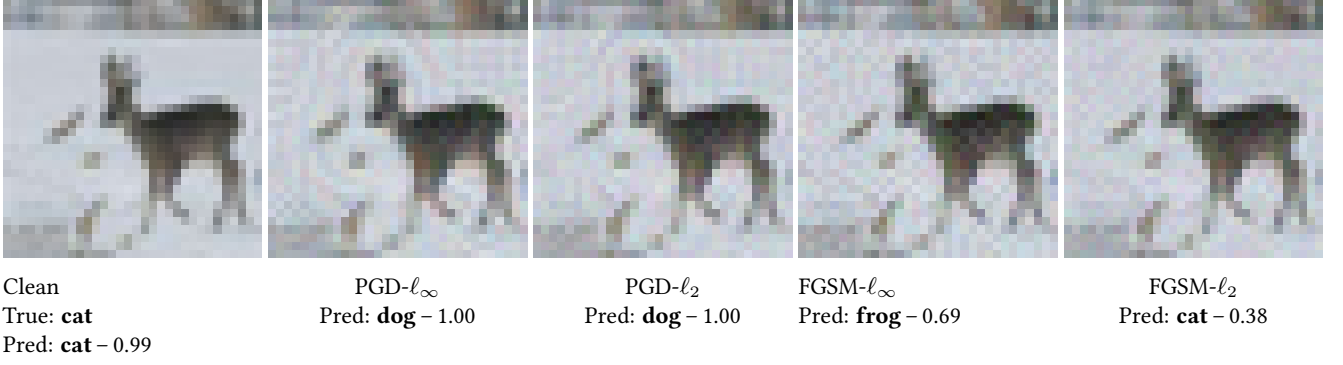
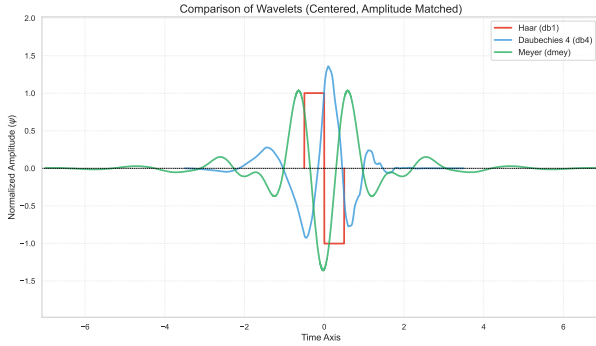


Figure 1: Effect of different adversarial attacks on a CIFAR-10 image. The clean image is classified correctly with high confidence, but PGD and FGSM perturbations (with standard budgets) are sufficient to induce misclassification. Note that PGD-based attacks transfer more strongly, producing high-confidence misclassifications, while FGSM- ℓ_2 induces a milder effect.



Examples of 1D wavelets (Daubechies 4, Haar, Meyer)

Besov regularity and entropy structure Wavelet coefficients provide a multiscale description whose decay encodes the smoothness of f . For $s > 0$, a function belongs to the Besov space $B_{p,q}^s$ when its wavelet coefficients $(\langle f, \psi_\lambda \rangle)_\lambda$ satisfy the scale-dependent summability condition

$$\sum_{j \geq 0} 2^{jsq} \left(\sum_{\lambda \in \Lambda_j} |\langle f, \psi_\lambda \rangle|^p \right)^{q/p} < \infty.$$

Higher regularity induces faster decay across scales, effectively reducing the number of significant coefficients. This property links directly to entropy bounds: functions with bounded Besov norm admit L^2 coverings of size $\varepsilon^{-d/s}$, yielding low metric entropy at fine resolution ε . Hence, Besov regularity provides a quantitative measure of information concentration across scales.

Feature extraction via Wavelet Scattering. The Wavelet Scattering Transform (WST) introduces a mathematically rigorous alternative to the initial learned layers of Convolutional Neural Networks (CNNs). Instead of learning filters from data, the WST relies on a cascade of fixed, pre-defined wavelet convolutions followed by a modulus non-linearity and a low-pass averaging. The main references are Oyallon et al. [2018], Mallat [1999, 2012].

Formally, let ψ_λ be a family of wavelets indexed by scale and orientation λ , and let ϕ_J be a low-pass averaging kernel at scale 2^J . The scattering coefficients of an input signal f are computed by cascading wavelet modulus transforms. The zeroth, first, and second-order coefficients are given by:

$$S_J f(x) = \left\{ f * \phi_J(x), |f * \psi_{\lambda_1}| * \phi_J(x), |f * \psi_{\lambda_1}| * \psi_{\lambda_2}| * \phi_J(x), \dots \right\} \quad (1)$$

By discarding the phase of the wavelet coefficients via the modulus operator $|\cdot|$, the WST gains translation invariance up to the scale 2^J , while recovering the high-frequency information lost in standard averaging pooling through the subsequent scattering orders.

Theoretical Stability. A key advantage of the WST over standard CNNs is its provable stability. Mallat [2012] demonstrated that the scattering operator does not increase the norm, and is Lipschitz continuous with respect to small diffeomorphisms. Formally, if L_τ is a deformation operator such that $L_\tau f(x) = f(x - \tau(x))$, then:

$$\|S_J(L_\tau f) - S_J f\| \leq C \|\nabla \tau\|_\infty \|f\|$$

This theoretical result ensures that small geometric distortions in the input space result in proportionally small changes in the feature space. Which is the desired property for attack-robust methods. This contrasts with standard CNNs, where instabilities in high-frequency components can lead to amplifying small input variations.

Robustness to Adversarial Attacks. This mathematical stability translates into empirical robustness against adversarial perturbations. While deep networks are prone to sensitivity to high-frequency noise (adversarial examples), the WST imposes a strong inductive bias through its fixed geometry. Since the filters are not learned, they cannot overfit to non-robust features. Recent works suggest that replacing the early layers of a network with scattering priors significantly

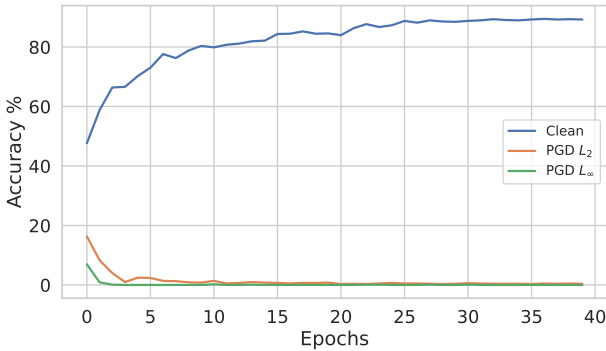
improves robustness to gradient-based attacks (FGSM, PGD, ...) and common corruptions, effectively acting as a regularizer that filters out non-semantic high-frequency noise Oyallon et al. [2018]. This property of filtering high frequency signal is the one which bring us to look at wavelets as the most promising perspective for methods robust to adversarial attacks.

Conclusion A model using WST would therefore be created using a differentiable wavelet like db4. The features extracted after WST would be fed into an MLP, which would be trained in a standard way based on the sparse representations extracted. This would be expected to result in better performance and greater robustness to attacks.

3 Experiments

3.1 Training for Clean Accuracy

We begin by training the Deep CNN model without any adversarial defense. With the full data augmentation pipeline described earlier, the model achieves a clean accuracy of approximately 89%. However, despite its positive effect on generalization, data augmentation alone offers almost no robustness against PGD adversaries. This confirms the well-known observation that standard training—even with heavy augmentation—does not inherently protect against adversarial perturbations.

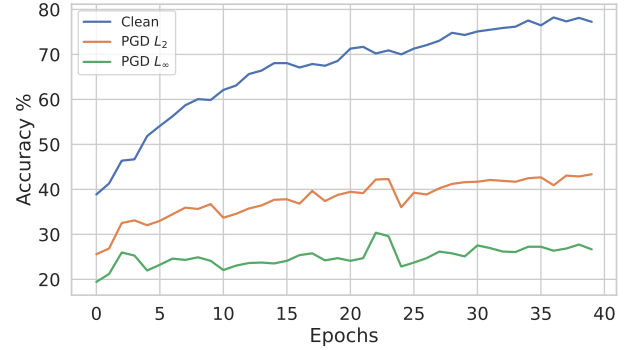


Clean and robust accuracies of the Deep CNN model trained without adversarial defenses.

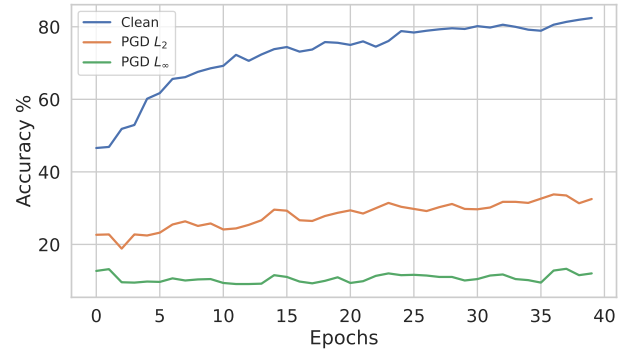
3.2 Adding Adversarial Training

We then incorporate adversarial training using different threat models. Our experiments show that training against the stronger ℓ_∞ adversary consistently yields models with higher robustness across both ℓ_∞ and ℓ_2 evaluations. Interestingly, while ℓ_∞ -trained models generalize fairly well to ℓ_2 attacks, the reverse is not always true: some models trained only against ℓ_2 perturbations perform significantly worse when evaluated under ℓ_∞ attacks. This asymmetry

highlights the importance of selecting a sufficiently strong adversarial objective during training.



FGSM ℓ_∞ Adversarial Training



FGSM ℓ_2 Adversarial Training

We should also note that some of the most aggressive data augmentation layers were removed from the pipeline during adversarial training for better training stability.

3.3 Adding TRADES Loss

The only modification in this experiment is the replacement of the standard cross-entropy loss with the TRADES loss, which makes the method both extremely efficient and straightforward to implement. Despite this minimal change, TRADES is grounded in a solid theoretical framework Zhang et al. [2019], unlike several heuristic approaches we explored. As a result, it yields a substantial improvement in robustness, increasing the robust ℓ_∞ accuracy from 35% to 41%. 2

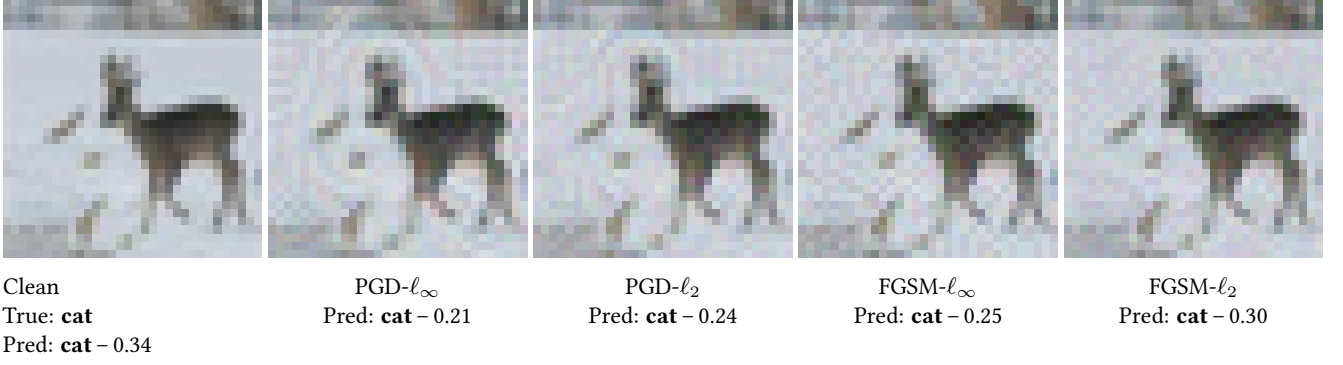
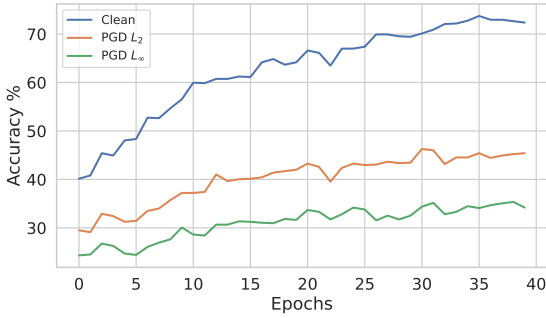
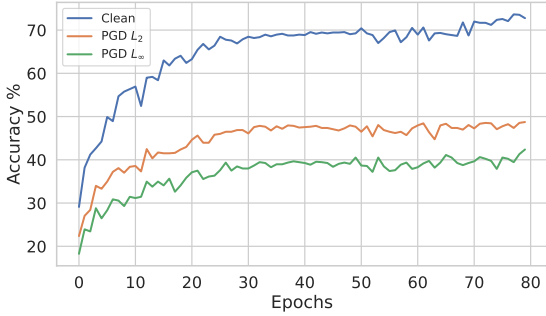


Figure 2: Final predictions with our most robust model. Despite reduced confidence on the clean image, the model remains unperturbed against all attacks.



10-iteration ℓ_∞ -PGD with CE loss



10-iteration ℓ_∞ -PGD with TRADES loss

3.4 Performances Table

The following table summarises the performance i accuracy of our main models on locally implemented metrics.

Method	Acc Nat	Acc PGD ℓ^2	Acc PGD ℓ^∞	Aggregated
FGSM ℓ^2	81	32	12.6	44.6
FGSM ℓ^∞	78	42	25.7	67.7
PGD ℓ^∞ CE	72	45	34	79.0
PGD ℓ^∞ TRADES	73	49	42	101
Wavelet	*	*	*	*

3.5 Wavelet Feature extraction

Unfortunately, by lack of time and some coding issues, we do not succeed to implement a fully working feature extractor using Meyer’s wavelet. Despite it’s highly promising expected performances, we will not comment the results linked to it. However it is the most remarkable perspective of work in this fiels, bringing frugality and robustness to attacks for signal classification methods.

4 Conclusion

In this project, we conducted a practical study of adversarial robustness on CIFAR-10, focusing on the effects of strong data augmentation, PGD-based adversarial training, and the TRADES objective. While extensive augmentation significantly improved clean accuracy, it provided virtually no robustness on its own, reaffirming the necessity of explicit adversarial defenses. Standard PGD training yielded robust yet unstable models, with performance strongly dependent on the choice of norm and hyperparameters. In contrast, TRADES offered a simple and theoretically grounded alternative, producing smoother optimization dynamics and noticeably higher robust accuracy with minimal implementation overhead. Altogether, our experiments illustrate the fundamental trade-off between accuracy and robustness, and highlight TRADES as an effective and practical method for improving adversarial resilience on small-scale datasets such as CIFAR-10.

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